Nikola Vojvodić

Full-Stack Developer

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PROFILE

Information Technology graduate (Faculty of Technical Sciences Čačak, 2026) who builds complete, working web applications in JavaScript and TypeScript. My projects range from a booking system with its own API and PostgreSQL database, through an offline-first Progressive Web App, to a real-time 3D product site, each built from scratch with attention to clean code, accessibility and automated testing, and each with a live demo and public source code.

SKILLS

Programming	JavaScript (ES2022+), TypeScript, HTML5, CSS3, Python
Frontend	React, Tailwind CSS, Progressive Web Apps (Service Worker, Cache API, Web App Manifest), responsive design, accessibility (WCAG 2.1), Three.js (WebGL), GSAP, i18n
Backend & data	Node.js, Express, REST APIs, PostgreSQL (migrations, constraints, indexes), Zod validation
Testing & tools	Vitest, Playwright, node:test, axe-core, Git, GitHub, GitHub Actions, npm, VS Code, Figma
Languages	Serbian (native), English (professional working proficiency)

PROJECTS

Rebound – full-stack court booking app — React · TypeScript · Node.js · PostgreSQL

- Booking site for a padel club concept: guests pick a court, a day and a time and get a booking code with a cancel link; staff sign in to see the day's schedule, drag bookings to move them and block courts for tournaments.
- REST API in Node.js and Express with plain SQL on PostgreSQL. Two bookings on the same court cannot overlap: the database itself checks it with an exclusion constraint, so it holds even when two people book in the same second.
- 16 tests in GitHub Actions, with the API tests running against a real PostgreSQL database. Frontend in React, TypeScript and Tailwind CSS; API hosted on Render, database on Neon.

  Live demo: nikolxdev.github.io/rebound/ · Code: github.com/nikolxdev/rebound

Knjigoteka – offline-first book discovery PWA — Vanilla JavaScript · Service Worker · Playwright

- Search app for the Open Library catalogue with a personal reading shelf (reading status, ratings, notes, reading statistics), JSON backup, Serbian/English UI and light/dark themes. Started as my bachelor thesis, graded 10/10.
- Service worker with three caching strategies (precache, network-first with timeout, capped cache-first) and a user-confirmed update flow; no framework or build step, strict Content-Security-Policy and XSS-safe rendering.
- 40 unit tests and 34 end-to-end tests on desktop and mobile viewports, including automated accessibility checks and a real offline run; CI pipeline on GitHub Actions.

  Live demo: nikolxdev.github.io/knjigoteka/ · Code: github.com/nikolxdev/knjigoteka

Morrow – 3D product site for a juice brand concept — Three.js · JavaScript · Playwright

- Real-time 3D cans built entirely in code (lathe geometry, canvas-drawn labels, instanced condensation drops) in a cover-flow row with floor reflections; clicking a can turns it around and opens its details.
- Lights-on intro with a matching still frame while the 3D loads, a saved shopping bag, a harvest calendar driven by today's date, shareable links per flavor, and rendering that pauses when the scene is off-screen.
- 8 Playwright tests on desktop and mobile in GitHub Actions, run with no retries so flaky tests stay visible.

  Live demo: nikolxdev.github.io/morrow/ · Code: github.com/nikolxdev/morrow

Last Ember – restaurant website concept — HTML · CSS · Vanilla JavaScript

- Website for a late-night burger bar: a burger that builds itself on scroll (with a reduced-motion fallback), a searchable menu with allergens and extras, a cart with delivery, promo codes and pickup slots, table booking and a live open/closed status; English/Serbian, no dependencies.

  Live demo: nikolxdev.github.io/last-ember/ · Code: github.com/nikolxdev/last-ember

EDUCATION

Bachelor's degree in Information Technology — 2022 – 2026

Faculty of Technical Sciences Čačak, University of Kragujevac

Thesis: *Development of a Progressive Web App (PWA) for Searching and Browsing Books Using Modern Client-Side Web Technologies* – grade 10/10

Technical School Čačak – Electronics Technician